Sabaq Live

A Real-Time Lecture Document Sharing, Student Response Collection, and AI-Assisted Question Generation Web Platform

The Islamia University of Bahawalpur

Final Year Project Proposal

Program: BSIT

Nature of Project: Development

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Area of Specialization: Web Development

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1. Introduction

Sabaq Live is a lightweight, mobile-friendly web platform designed to support classroom teaching. A teacher signs in, creates a session, uploads and presents a PDF document such as lecture notes, slides, or an exam paper, and shares a unique join link or code with students. Students join through a browser and view the same document that the teacher is presenting. When the teacher changes the PDF page or switches to another document, students' views update automatically in near real time.

During a session, the teacher can send questions to the class. Students submit responses through the same platform, and the teacher can review submissions together with each student's name and registration number. The AI component will analyze text extracted from an uploaded PDF and suggest two types of study questions: multiple-choice questions (MCQs) and important short-answer questions. The teacher can review, edit, select, and share suitable questions with students.

The aim is to combine live document presentation, student response collection, and AI-assisted question preparation in one platform, reducing the need to switch between separate tools.

2. User Roles

Teacher: Signs in with an account, creates sessions, uploads and presents PDFs, sets a session password, shares a join link or code, sends questions, reviews student responses, and generates/reviews AI-suggested MCQs and important short-answer questions from lecture material.

Student: Joins a session using the join link or code and session password, views the live PDF, and submits responses to teacher questions or selected quiz questions. Students do not need separate accounts; they enter their name when joining. The system stores the name with submitted responses and prevents the same name from joining the same session twice at the same time.

3. Problem Statement

Teachers often rely on several separate tools to present lecture documents, ask questions, and collect student responses. This can interrupt lessons, create confusion, and make it difficult to ensure that all students are viewing the same page of a document. Separately shared files do not automatically stay synchronized with the teacher's presentation. Preparing quiz questions and identifying important short questions manually from lecture PDFs can also take additional time.

Sabaq Live addresses these issues by combining synchronized PDF presentation, in-session questions and responses, and an AI-assisted feature that drafts MCQs and important short-answer questions from uploaded lecture material. Generated questions remain suggestions for the teacher to review and approve.

4. Project Goal and Objectives

Goal: To develop a responsive web platform for real-time PDF presentation and student response collection, with an AI-assisted feature for suggesting MCQs and important short-answer questions from lecture documents.

- Provide a login page and role-based access for teachers and students.
- Allow teachers to create sessions and share a unique join link or code with students.
- Display the teacher's current PDF page on students' screens and update it in near real time when the teacher changes pages or documents.
- Allow teachers to send questions and collect student responses in the same session.
- Store and display student names and their submitted responses for teacher review.
- Extract text from uploaded PDF documents and use an AI service to suggest multiple-choice questions (MCQs).
- Use the extracted PDF content to suggest important short-answer questions for revision and classroom discussion.

- Allow teachers to review, edit, select, and share the generated questions with students.
- Provide a responsive interface that works on desktop and mobile browsers.

5. Application Architecture, Technology Stack, Tools and Hardware

Languages: Python (backend), HTML, CSS, and JavaScript (frontend).

Frameworks and Libraries: Flask for the Python web application; Flask-SocketIO for real-time updates between teacher and students; PDF.js for displaying PDFs in the browser; and a PDF text-extraction library such as pypdf for reading PDF text.

AI Component — PDF-to-Question Suggestions: After the teacher uploads a lecture PDF, the system extracts its text and sends relevant content to an AI service. The AI suggests (1) MCQs with answer options and a correct answer, and (2) important short-answer questions based on key concepts in the document. The teacher can review, edit, remove, and select questions before sharing them with students. The AI does not automatically publish questions without teacher approval.

Database: SQLite, a lightweight file-based database that does not require a separate database server. Uploaded PDF files will be stored on the local server's file system, with their paths and session information stored in the database.

Architecture: Client-server architecture using Flask routes and templates, SQLite for data storage, and Flask-SocketIO for real-time communication.

Development and Demonstration: The application will be developed and run locally on the student's laptop for testing and demonstration. Cloud deployment is not required for the planned demonstration.

Tools: Visual Studio Code, Python 3, Git, Google Chrome, and draw.io for diagrams.

Hardware: A laptop or computer with a Core i5 processor, 8 GB RAM, and internet access for package installation and AI API requests.

6. Scope and Limitations

The project focuses on classroom sessions, synchronized PDF viewing, teacher questions, student response collection, and AI-assisted generation of draft MCQs and important short-answer questions from lecture PDFs. AI-generated questions may require correction and approval by the teacher. The AI feature depends on access to an external AI API, so internet access may be required and provider usage limits or fees may apply. The core application is intended to run locally on the student's laptop; public cloud hosting is outside the planned scope.

7. Expected Outcome

A working, responsive web application that supports teacher-led live PDF presentation, student response collection, and AI-assisted suggestions for MCQs and important short-answer questions based on lecture PDFs. The project deliverables will also include Software Requirements Specification (SRS) and Software Design Description (SDD) documents.

Supervisor Acceptance Signature: _____

Head of Department (HoD): _____